



# Supply Chain Security, Procurement Continuity

Presenting to: BC Hydro

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Presented by:



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**Objective**

Identify the category most vulnerable to supply chain disruption and consider strategic approaches for the short and long term

**Problems**

**Immediate** procurement  
**vulnerability** across all  
categories

Category vulnerability from  
**USA imports**

Approach a 25% USA tariff in  
both **short** and **long term**

**Strategy****Smart Sourcing****Resilience**

Target low buffer, low-risk  
categories

**Value**

Diversify portfolio for future  
growth and security

**Key  
Takeaway**

Leveraging global vendors in the short term will protect vulnerable categories, while long term partnerships in Europe will expedite procurement delivery





## Data analysis break down for efforts focused on highly important, highly vulnerable categories

### Methodology

Low Risk Tolerance



**Prioritizing low risk tolerance** categories as an operational necessity

Buffer Stock



Buffer stock in days as a **proxy for vulnerability**  
(=Total days of supply – Avg. Lead time)

Lead Time



**Lead time** as the main decision criterion for supply chain analysis **in highly vulnerable categories**

Performance  
& Tariffs



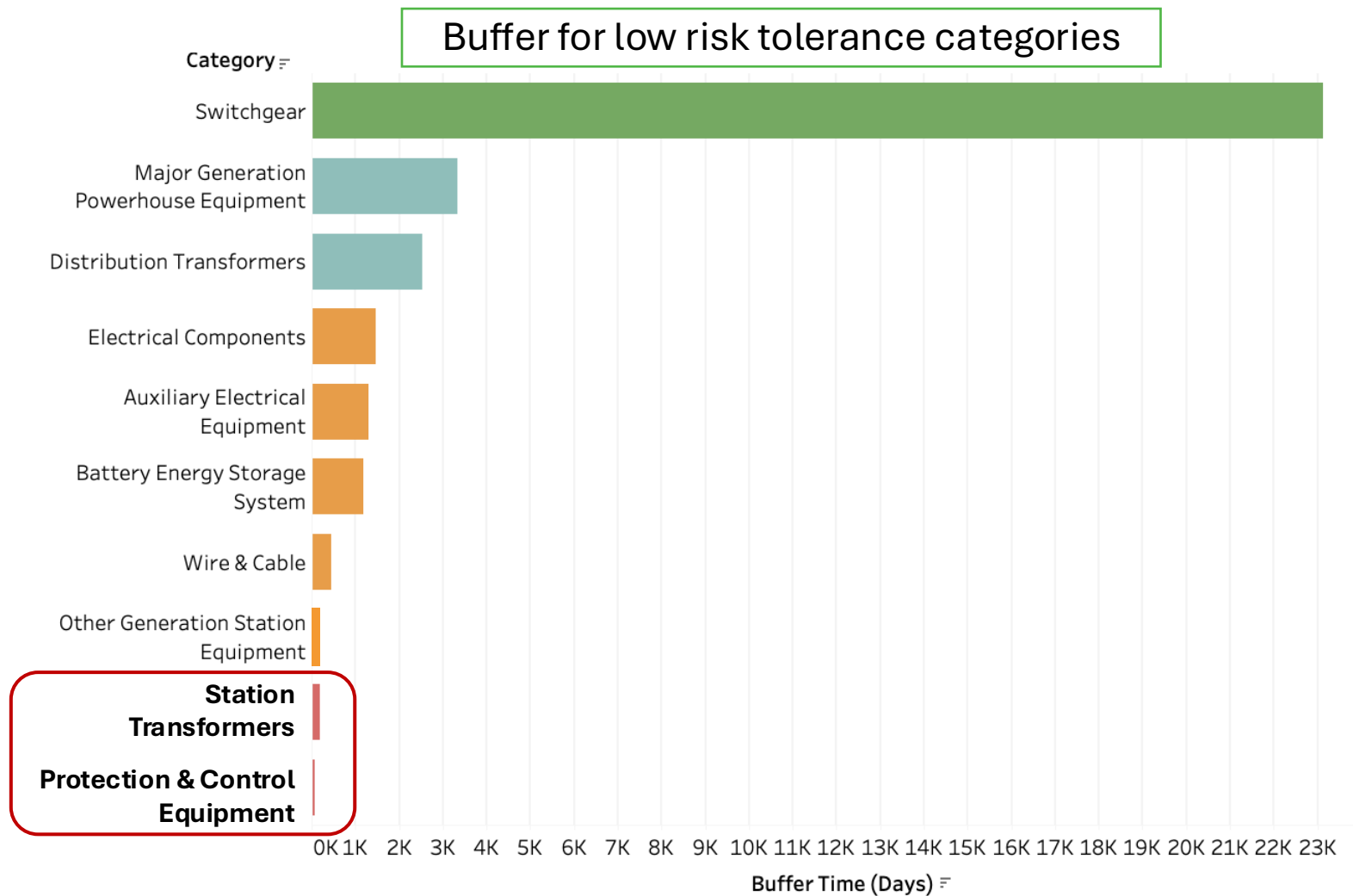
Factoring **vendor performance (1-5)** and **tariff rates**



Buffer stock measure vendors active inventory, frequency of use, and average lead time

### Buffer stock accounts for

- Vendors active inventory
- Frequency of use
- Average lead time





Two low risk tolerance categories show vulnerability from low buffer and long lead times

### Buffer stock accounts for

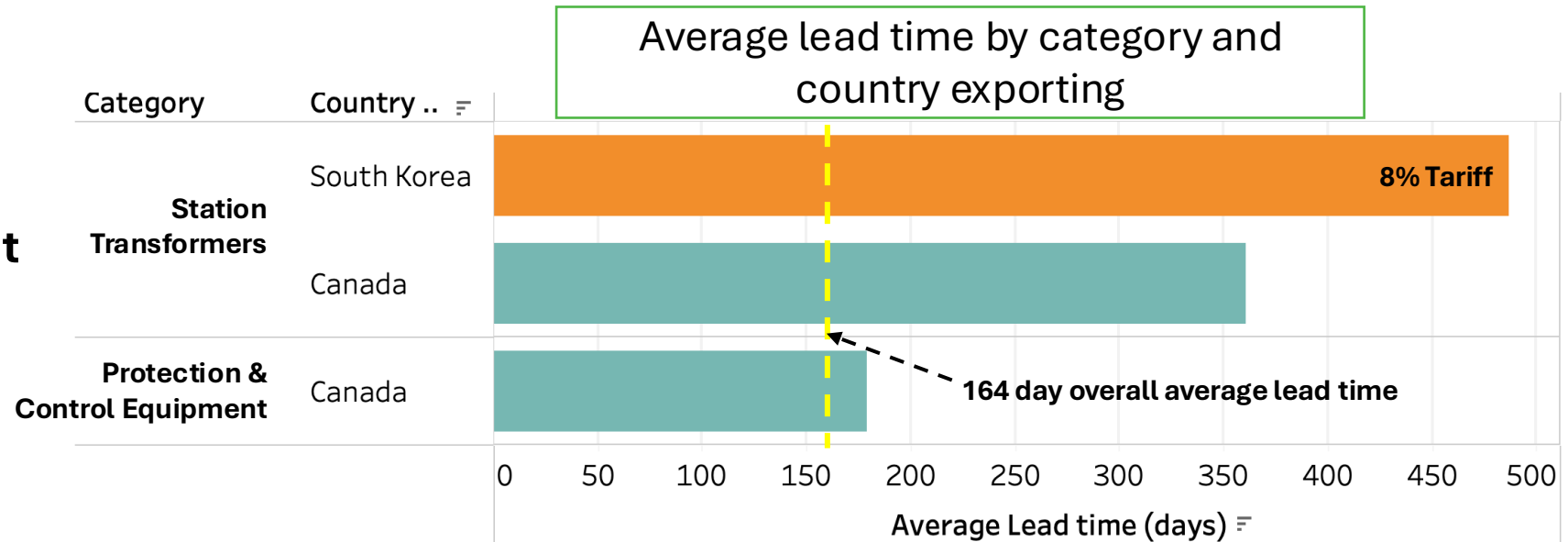
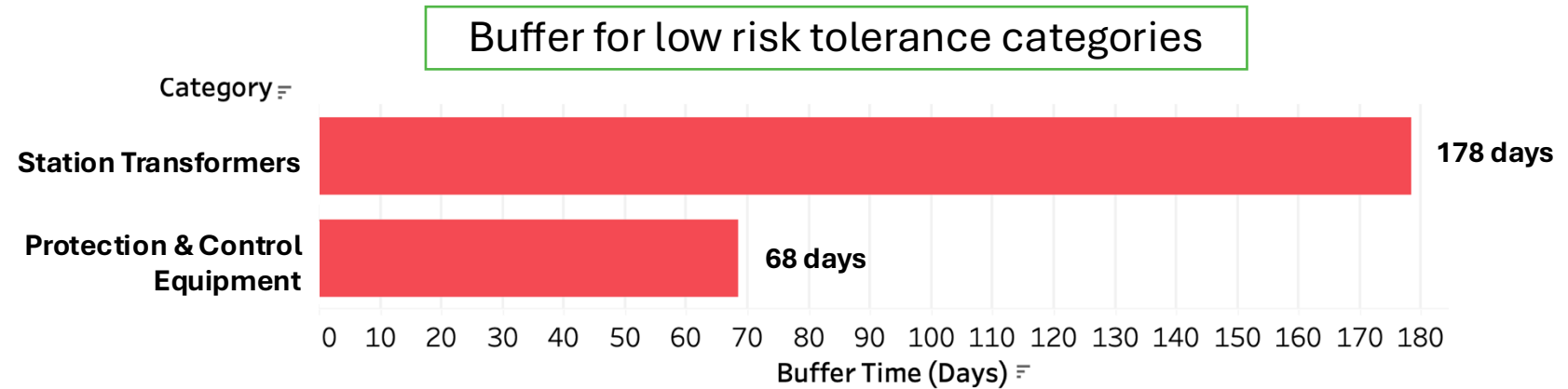
- Vendors active inventory
- Frequency of use
- Average lead time

### Station Transformers

- Long Lead Times
- Subject to 8% Korean Tariff

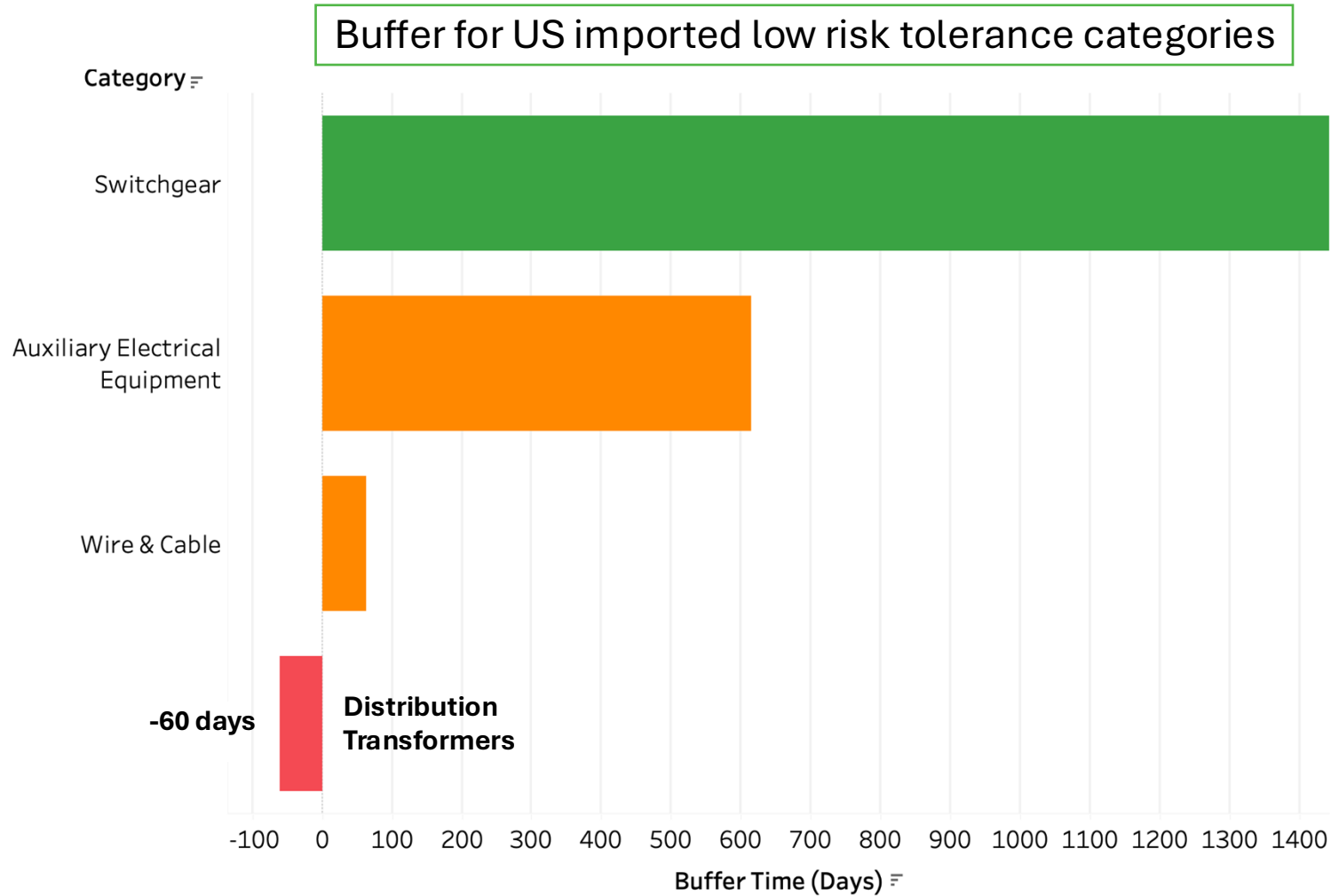
### Protection & Control Equipment

- Lowest buffer stock
- Reliant on **sole vendor**
- Low safety stock (12)





## Distribution transformer inventory will be empty for 2 months without a vendor change

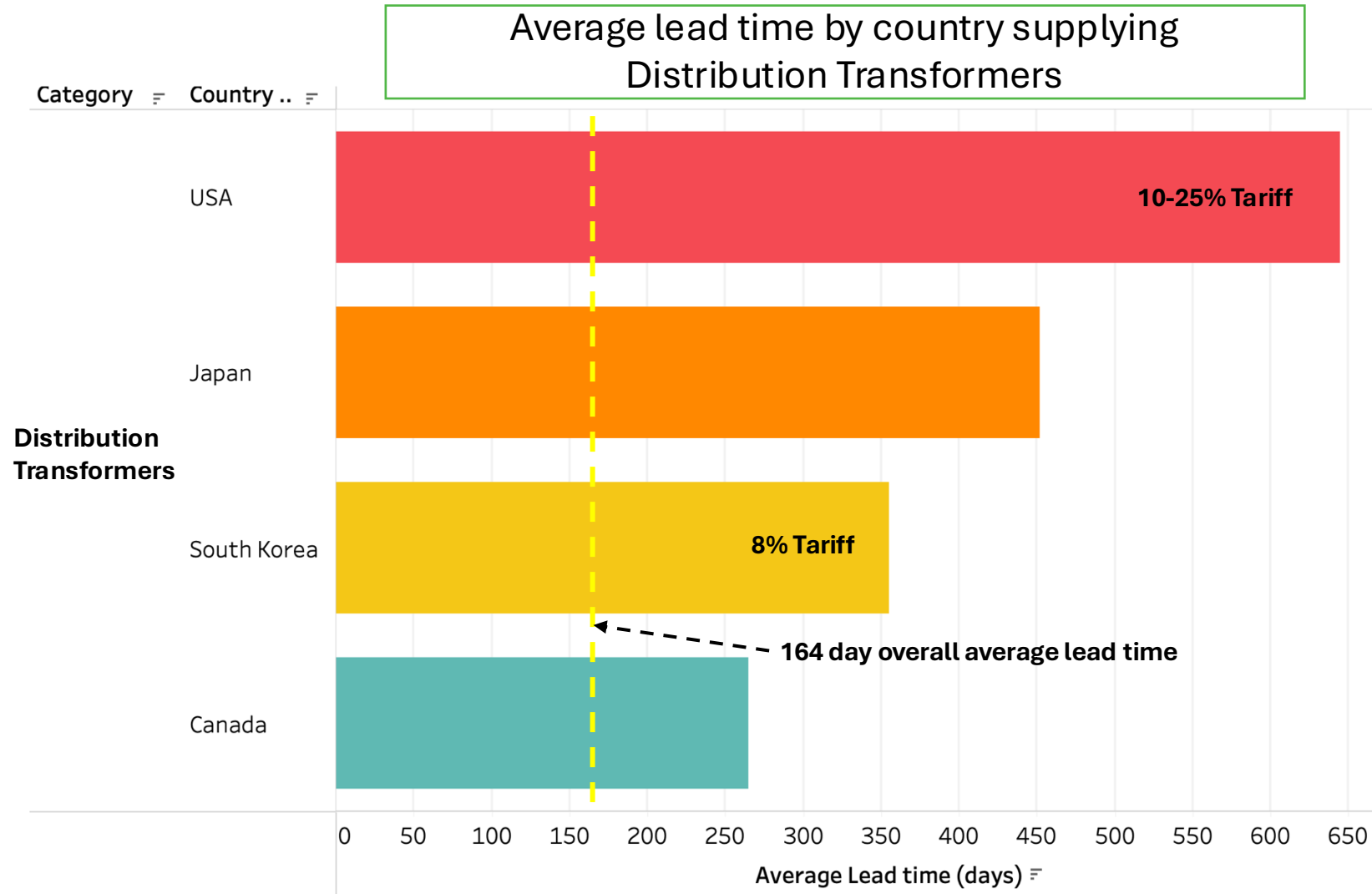


### Distribution Transformers

- -60 day buffer means **no stock** for 2 months
- Large standard deviation +/- 40% of average lead time



## BC Hydro has vendors options around the world for Distribution Transformers



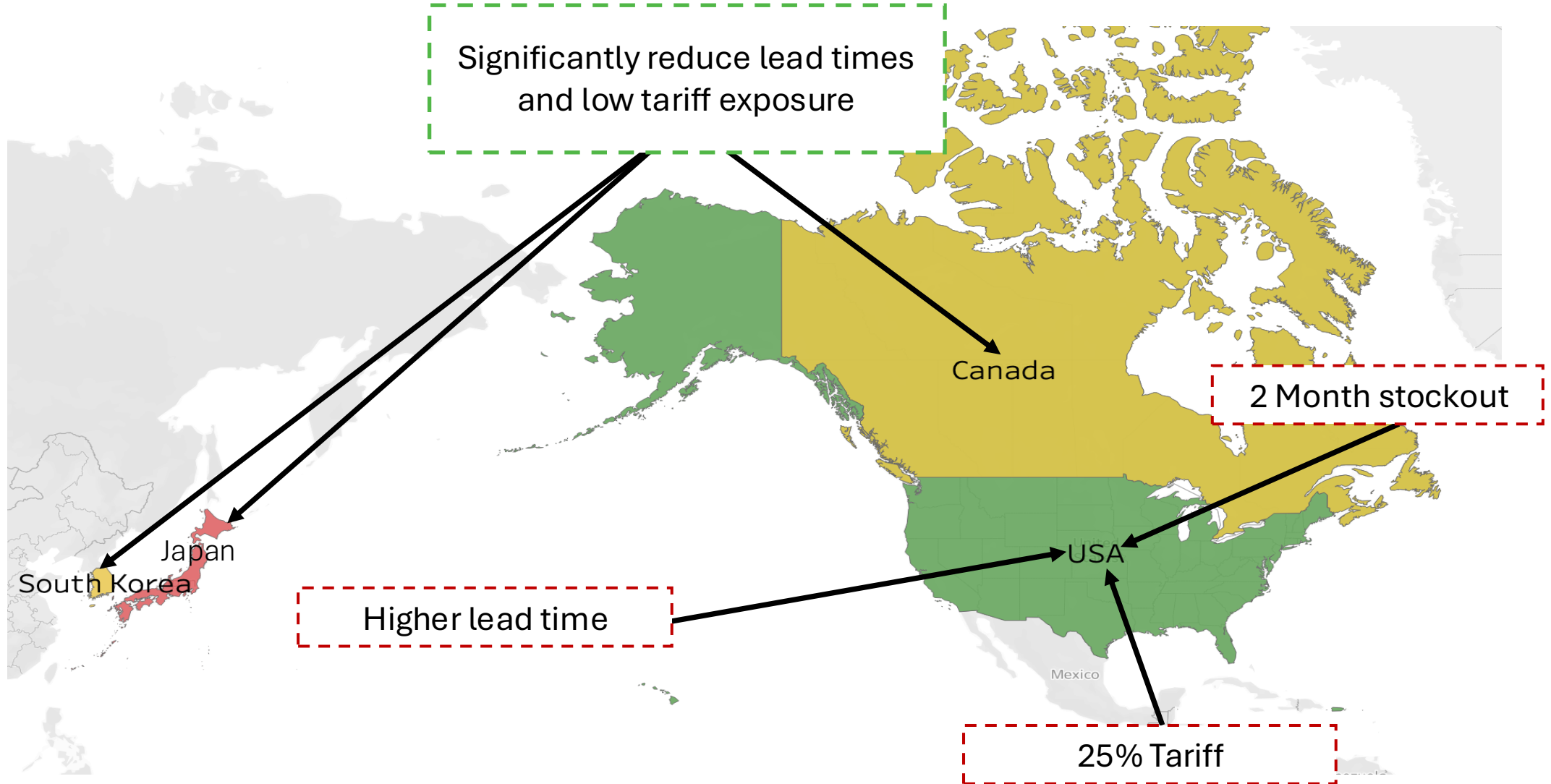
### Distribution Transformers

- -60 day buffer means **no stock** for 2 months
- Large standard deviation +/- 40% of average lead time

### Strategic Opportunity

- Japan, South Korea, and Canada offer **lower tariff rates and lead times**

# Transition distribution transformer procurement to South Korea and Canada to minimize lead times and tariff exposure



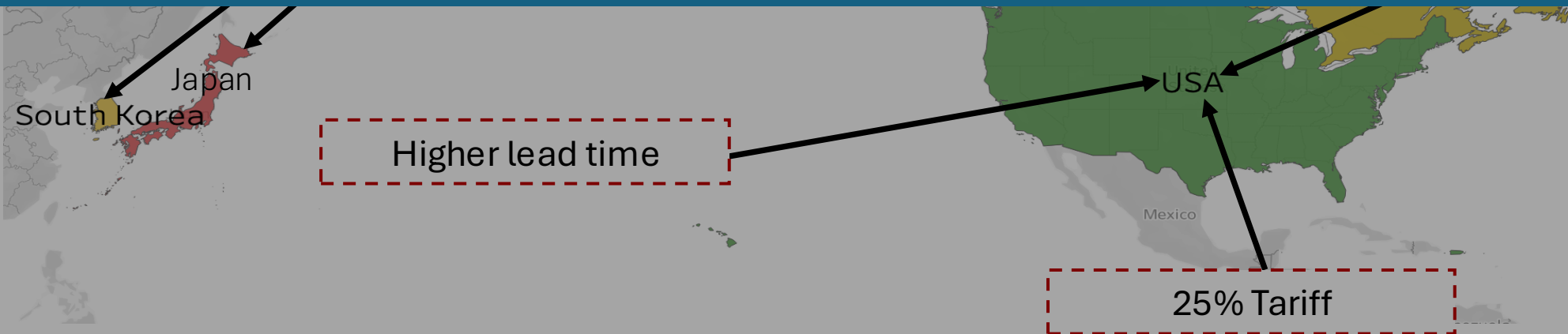


## Transition distribution transformer procurement to South Korea and Canada to minimize lead times and tariff exposure

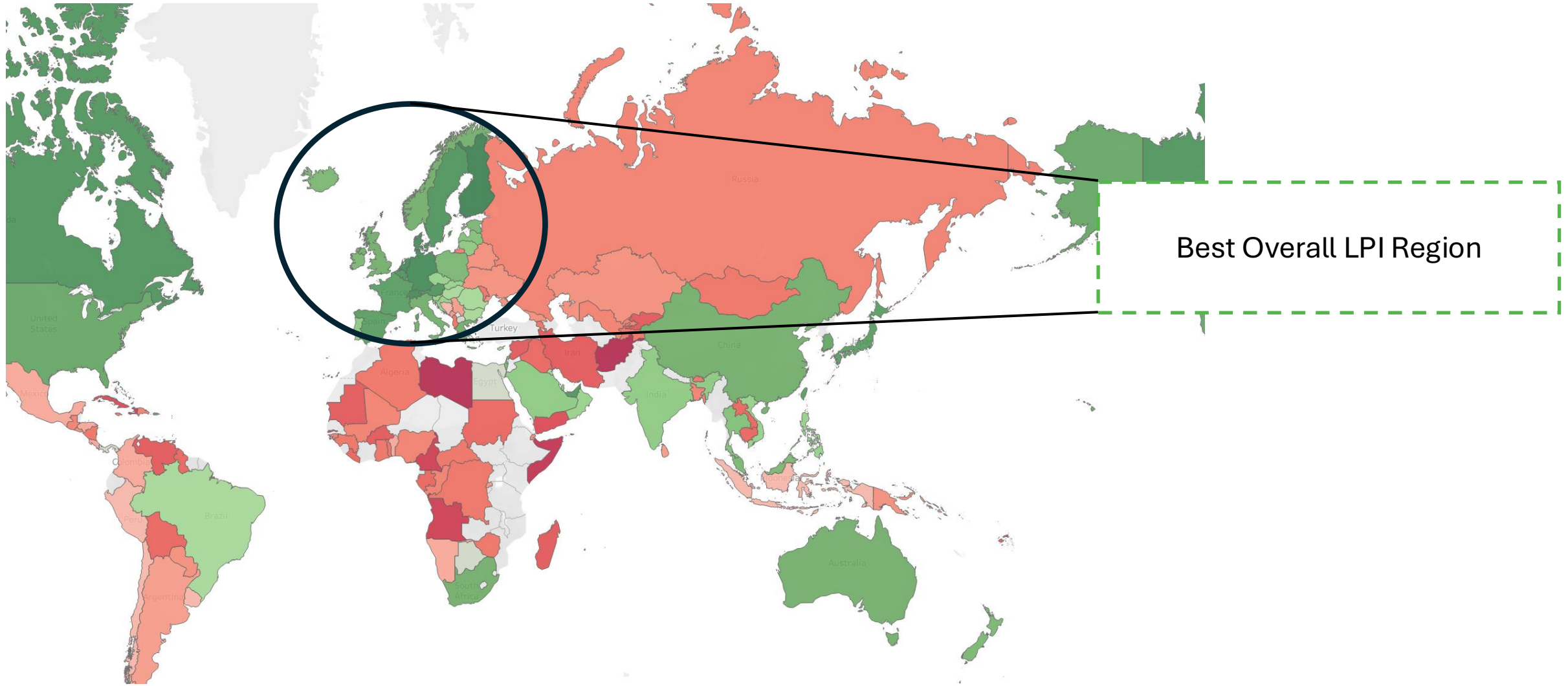


Significantly reduce lead times  
and low tariff exposure

Prioritize sourcing distribution transformers from **South Korea and Canada** instead of the U.S. to significantly reduce lead times and tariff exposure

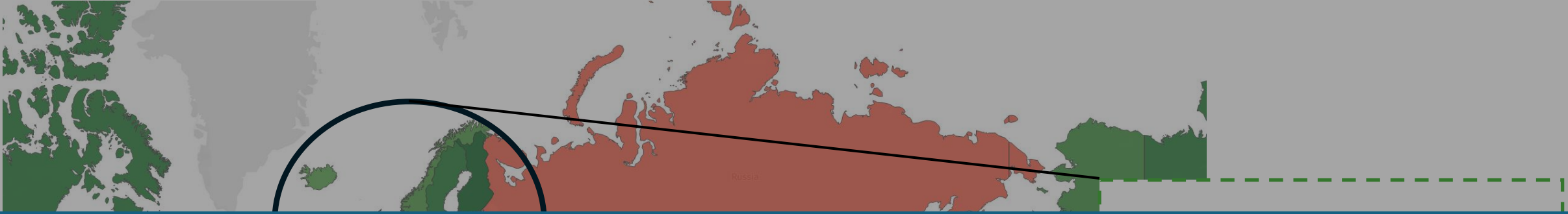


## Considering potential alternatives based on LPI

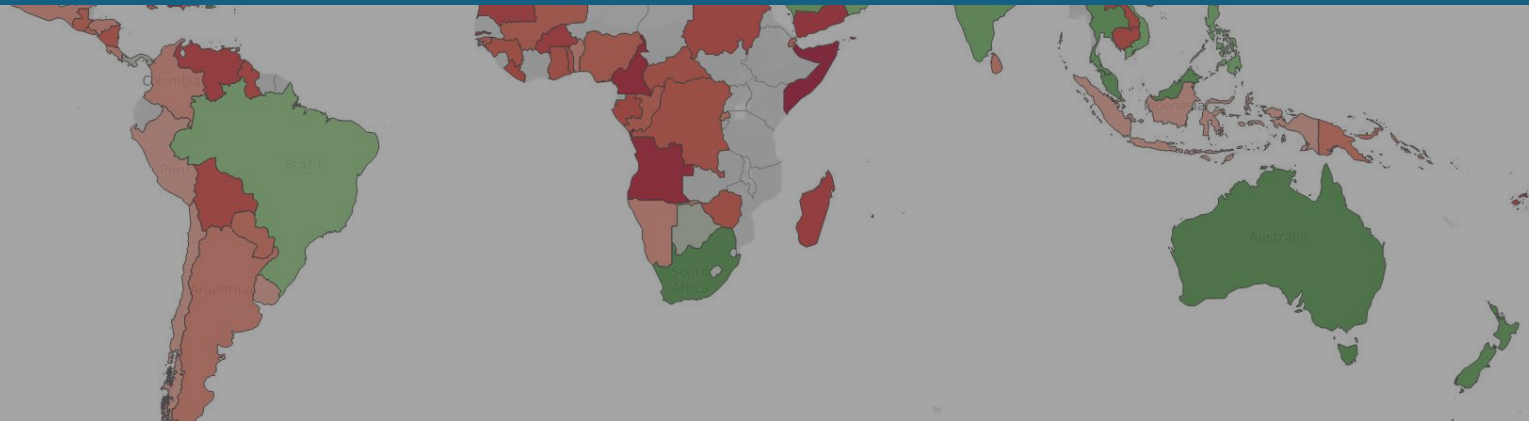




## Considering potential alternatives based on LPI



Urgent need to diversify into Europe with potential increasing USA tariff





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# Appendix



## Short-term Strategy

**KEY ISSUE:**

- **Distribution transformers** have a buffer time of ~60 days, worst case there could be no on-hand stock for **two months**
- **Vendor Pool:** Available suppliers are in the USA, Japan, South Korea, and Canada
- Although USA vendors have the best performance, **longer lead times and 25% tariffs** is why we need alternatives

**High lead time + High tariffs + Negative buffer = Significant stockout and Cost risk**

**Result:**

Two-Month Stockout Risk  
Protracted Replenishment  
Escalated Procurement Costs



Prioritize sourcing distribution transformers from South Korea and Canada instead of the U.S. to significantly reduce lead times and tariff exposure

